

Atherosclerosis, Intracranial Aneurysms, and Intermediate Biomarkers: Real-World Observational and Mendelian Randomization Research

Wei Liu, Zhaoxu Zheng, Chenglong Liu, Yuanren Zhai, Shuang Wang, Liangran Huang, Rong Wang, Yan Zhang, Peicong Ge , Dong Zhang 

Department of Neurosurgery, Beijing Tiantan Hospital, Capital Medical University, Beijing, China • China National Clinical Research Center for Neurological Diseases, Beijing, China • Center of Stroke, Beijing Institute for Brain Disorders, Beijing, China • Department of Neurosurgery, Beijing Hospital, National Center of Gerontology, Beijing, China • Institute of Geriatric Medicine, Chinese Academy of Medical Sciences, Beijing, China

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

eLife Assessment

This is **valuable** work with theoretical implications for possible mediation by MMP12 in the link between atherosclerosis and intracranial aneurysms, using Mendelian Randomization for causal inference. Additional analysis would be required to verify the claims, which currently have **incomplete** support in terms of the strength of evidence. Given that most of the identified causal associations do not hold after correcting for multiple tests, the conclusions should be carefully reviewed in order to be fully supported by the results.

<https://doi.org/10.7554/eLife.106521.1.sa3>

Abstract

Background

The causal relationship between atherosclerosis and the development of aneurysms remains controversial. This study aims to analyze the genetic predictive association between atherosclerosis and intracranial aneurysms, as well as to explore intermediate biomarkers that may mediate the causal relationship between these two conditions.

Methods

This study utilized head imaging data for a cross-sectional analysis, investigating the prevalence of cerebral atherosclerosis and intracranial aneurysms in 13,739 patients. We performed Mendelian randomization analysis on two samples to explore the genetic predictive association between atherosclerosis and intracranial aneurysms. Furthermore, we examined the role of the matrix metalloproteinase family as an intermediary biomarker in the causal relationship between these two conditions and validated biomarker in serum and in situ samples from clinical patients.

Results

Cross-sectional study revealed that patients with intracranial aneurysms exhibited a higher prevalence of cerebral atherosclerosis compared to controls (14.155% vs. 7.069%, $P < 0.001$). Two-sample Mendelian randomization analysis demonstrated that genetically predicted peripheral atherosclerosis significantly increased the risk of intracranial aneurysms, encompassing both unruptured ($OR_{IVW} = 1.711$, $P_{IVW} = 0.002$) and ruptured intracranial aneurysms ($OR_{IVW} = 1.533$, $P_{IVW} = 9.41 \times 10^{-4}$). Mendelian randomization further indicated that peripheral atherosclerosis leads to elevated circulating MMP12 levels ($\beta_{IVW} = 0.083$, $P_{IVW} = 0.008$), which were subsequently associated with increased risks of unruptured ($OR_{IVW} = 1.136$, $P_{IVW} = 0.010$) and ruptured intracranial aneurysms ($OR_{simple-mode} = 1.217$, $P_{simple-mode} = 0.038$). ELISA assays confirmed elevated MMP12 levels in the plasma of patients with carotid artery stenosis and intracranial aneurysms. In situ analyses revealed upregulated MMP12 expression in both plaques and intracranial aneurysm tissues.

Conclusion

Our research clarifies the causal relationship between atherosclerosis and intracranial aneurysms, identifies and validates MMP12 as a key biomarker, and provides new methods and evidence to explain the association between these two diseases.

What is already known on this topic

Previous studies have suggested a significant association between atherosclerosis and the development of aneurysms.

What this study adds

This study provides robust evidence and causal inferences regarding this relationship, identifying intermediate biomarkers of the diseases.

How this study might affect research, practice or policy

These findings suggest that preventing atherosclerosis and controlling MMP12 levels may help reduce the risk of intracranial aneurysms.

Background

Intracranial aneurysm, a common cerebrovascular disease, typically manifests as an enlargement or protrusion of cerebral blood vessels. Research indicates a prevalence of 1%-2% in the general population¹, with a cross-sectional study in Shanghai revealing a 7% detection rate in adults aged 35 to 75 years through comprehensive brain magnetic resonance angiography (MRA) screening². Aneurysm rupture, which often leads to nontraumatic subarachnoid hemorrhage with high mortality rates, is primarily attributed to inflammation, matrix degradation, thrombosis, and hemodynamic factors according to animal model studies^{3,4}. Risk factors for intracranial aneurysms include smoking, hypertension, substance abuse, hereditary connective tissue disorders, histopathies, and polycystic kidney disease⁵.

Both atherosclerosis and aneurysms are complex diseases characterized by inflammatory cell infiltration, matrix degradation, and arterial wall thrombosis. A case-control study found that patients with risk factors for atherosclerosis are more likely to develop middle cerebral artery aneurysms⁶. Furthermore, a retrospective cross-sectional study observed that unruptured intracranial aneurysm patients with atherosclerotic plaques in the middle cerebral artery exhibited more pronounced aneurysm wall enhancement, suggesting a potential association between the presence of plaques and intracranial aneurysm characteristics⁷. A study using fluid-structure interaction analysis to investigate the interaction between blood and vascular tissue caused by atherosclerotic occlusion found that hemodynamic changes induced by atherosclerosis increase the pressure on the intracranial artery wall, thereby elevating the risk of arterial tissue remodeling and intracranial aneurysm formation⁸. However, the relationship between atherosclerosis and the development of intracranial aneurysms remains complex and is not fully understood, with ongoing debate about whether they share common risk factors or whether there is a direct causal link between them.

The potential comorbidity and causal relationship between atherosclerosis and intracranial aneurysms have not been definitively established. This study aimed to investigate the prevalence of intracranial aneurysms and atherosclerosis based on a real-world cross-sectional study and to utilize Mendelian randomization to explore the possible causal relationships and common causative factors between these two diseases. Finally, the common pathogenic biomarkers identified through Mendelian randomization were validated in clinical samples.

Methods

1. Study design

This study explores the potential association between atherosclerosis and intracranial aneurysms, divided into four sections based on the progression of the analysis: real-world cross-sectional study, Mendelian randomization analysis, plasma validation, and in situ specimen validation. The detailed arrangement is shown in **Figure 1**. This study adhered to the guidelines outlined in the

Declaration of Helsinki and received approval from the Ethics Committee of Beijing Tiantan Hospital, which is affiliated with Capital Medical University (Ethics approval number: KY2022-051-02). Prior to enrollment, all participants provided informed consent for the collection of clinical information and imaging data.

2. Real-world cross-sectional study

The study population consisted of patients who underwent head Computed tomography angiography (CTA) or MRA examinations at Beijing Tiantan Hospital between January 1, 2023, and August 31, 2023. The inclusion criteria were as follows: (1) Head CTA or MRA examination with a diagnosis including the term ‘plaque’. (2) Head CTA or MRA examination with a description including the terms ‘stenosis’ or ‘occlusion’. Patients who had undergone multiple examinations were excluded, and only the results of their first examination were considered. The final sample size for the cross-sectional study was 13,739, and data on sex, age, and intracranial aneurysm incidence were collected. Intracranial aneurysm diagnosis was based on head CTA or MRA examinations that included the term ‘aneurysm’.

3. Mendelian Randomization analysis

We performed two-sample MR to investigate three sets of associations: (1) the association between genetic susceptibility to atherosclerosis and intracranial aneurysms; (2) the association between genetic susceptibility to atherosclerosis and intracranial aneurysms with circulating matrix metalloproteinase family levels; and (3) the genetically predicted association of circulating matrix metalloproteinase family levels with the risk of atherosclerosis and intracranial aneurysms.

We analyzed the causal relationship between atherosclerosis and intracranial aneurysms, considering the various subtypes of each disease. Intracranial aneurysms were classified as either unruptured or ruptured, while atherosclerosis was categorized into four subtypes based on location: cerebral atherosclerosis, coronary atherosclerosis, peripheral atherosclerosis and other atherosclerosis. Among them, peripheral atherosclerosis refers to atherosclerosis of the limbs. Other atherosclerosis refers to atherosclerosis occurring in all locations except the cerebral arteries, coronary arteries, and peripheral arteries of the limbs.

To investigate the potential role of plasma MMPs in the relationship between these two diseases, we selected 12 MMPs for analysis using a two-way MR approach. Our objective was to identify plasma MMPs associated with or induced by intracranial aneurysms and atherosclerosis and to evaluate their potential role in mediating the causal relationship between these two diseases.

4. Instrument Selection

The sources of genetic instrumental variables are presented in [Table 2](#). We performed two-sample MR analysis and two-way MR analysis using genetic instrumental variables with p values less than 5×10^{-9} for each exposure. If the genetic instrumental variables could not be identified with this p value, we adjusted the threshold to less than 5×10^{-6} . We excluded associated SNPs ($r^2 > 0.1$) by retaining the SNP with the smallest p value associated with the SNP exposure association. To avoid weak instrument bias, we included only SNPs with an F-statistic greater than 10^9 .

5. Sensitivity analysis

We performed sensitivity analyses using the MR-Egger, weighted median, and MR Pleiotropic Residual Sum and Outlier (MR-PRESSO) methods to evaluate robustness and assess horizontal pleiotropic effects when more than two instrumental variables were available. MRLEgger analysis estimated the pleiotropic effect of the instrumental variable, with nonzero intercepts indicating bias in the inverse variance weighted (IVW) estimates. The weighted median approach provides a causal estimate by considering the nullity of multiple genetic variants or the presence of

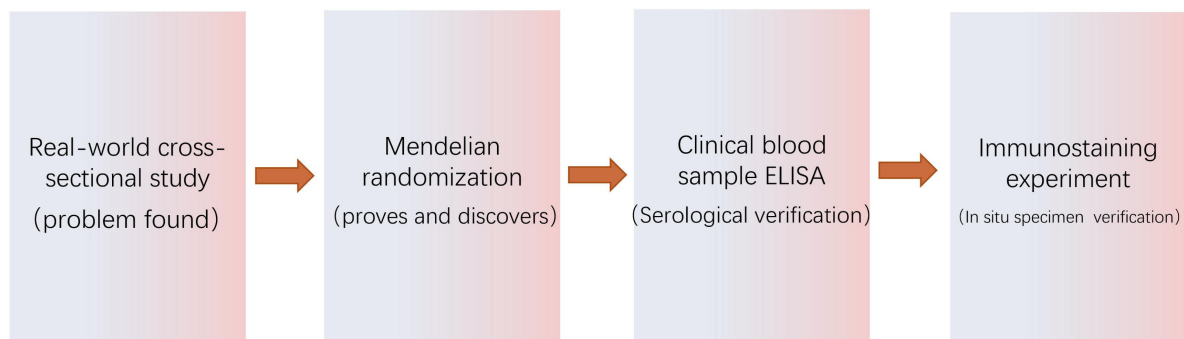


Figure 1.

Research arrangement.

pleiotropic effects. MR-PRESSO used a global test to detect horizontal pleiotropy and, if necessary, corrected for potential pleiotropic outliers by removing them¹⁰. The heterogeneity of the IVW estimates was evaluated using Cochran's Q test¹¹.

We performed bidirectional MR analyses to further explore the directionality of potential reciprocal relationships between diseases and between biomarkers and diseases. To correct for multiple testing, we used the Bonferroni method. In the MR analysis of the association between atherosclerosis and intracranial aneurysms, we used a p value threshold of $0.05/8 = 6.25 \times 10^{-3}$. In the two-way MR analysis of MMPs levels and diseases of interest, we considered 72 unique associations with a p value threshold of $0.05/72 = 6.94 \times 10^{-4}$. P values below the threshold after multiple comparisons were considered strong evidence of association, while those between the threshold and 0.05 were considered suggestive of associations. All analyses were two-sided and performed using the TwoSampleMR (version 1.0.3) and MRPRESSO (version 6.3) packages in R software (version 4.2.1). The report adheres to the STROBE-MR statement.

6. Plasma ELISA and in situ specimen validation

Detailed materials and methods for serum ELISA and in situ specimen validation are provided in the supplementary materials.

7. Statistical analysis

All analyses were performed using SPSS (version 25) and R (version 4.1.2). Categorical variables are presented as counts (percentages), and comparisons of proportions were conducted using the Pearson chi-square test. Continuous variables are described as either the mean $\pm$ standard deviation or median and interquartile range, depending on their distribution, which was assessed by the Kolmogorov-Lsmirnov test. All the statistical tests were two-sided.

Results

Cross-sectional study

This cross-sectional study involved a total of 13,739 participants, comprising 2,946 patients with intracranial aneurysms and 1,180 patients with cerebral artery atherosclerosis. The average age of the participants was 59 years, with a male population of 48%. Age did not differ significantly between patients with and without intracranial aneurysms. However, a greater proportion of patients with aneurysms were female (60.692%) than male patients (39.308%). The sex distribution varied between the groups. Additionally, patients with intracranial aneurysms exhibited a greater prevalence of cerebral artery atherosclerosis than did those without intracranial aneurysms (14.155% vs. 7.069%, $P < 0.001$). The original data of the cross-sectional studies are shown in eTable 1, and the baseline data are shown in **Table 1**. Even after adjusting for age and sex, multivariate regression analysis indicated that cerebral artery atherosclerosis remained an association with intracranial aneurysm (OR=2.439, 95% CI 2.139-2.781; $P < 0.001$).

Two-sample Mendelian randomization analysis (atherosclerosis and intracranial aneurysm)

Using double-sample MR analysis, we investigated the genetically predicted relationship between atherosclerosis and intracranial aneurysm, with atherosclerosis as the exposure and intracranial aneurysm as the outcome. **Figure 2** presents the results of eight consecutive analyses with P values less than 0.05. After adjusting for multiple comparisons, we found that genetically predicted peripheral atherosclerosis increased the risk of intracranial aneurysms, including unruptured intracranial aneurysms (OR_{IVW} = 1.711; 95% CI 1.208-2.423; $P_{IVW} = 0.002$) and

Table 1.

Cross-sectional study baseline table.

Variables	All patients (13739)	With aneurysm (n=2946)	Without aneurysm (n=10793)	P value
age	59.560 ± 10.275	59.494 ± 10.417	59.817 ± 9.738	0.131
sex/male	6588(48.000)	1158 (39.308)	5430 (50.310)	<0.001
Atherosclerosis	1180(8.600)	417 (14.155%)	763 (7.069%)	<0.001

Table 2.

Sources of Mendelian randomization genetic instruments.

Trials	Cases (N)	Controls (N)	Sample size (N)	Year of publication	No. of SNPs	Population	Source
Cerebral aneurysm, nonruptured	992	217,800	218,792	2021	16,380,466	European	FinnGen GWAS
Cerebral aneurysm, SAH	2,127	216,665	218,792	2021	16,380,466	European	FinnGen GWAS
Coronary atherosclerosis	23,363	187,840	211,203	2021	16,380,447	European	FinnGen GWAS
Cerebral atherosclerosis	104	218,688	218,792	2021	16,380,466	European	FinnGen GWAS
Peripheral atherosclerosis	6,631	162,201	168,832	2021	16,380,247	European	FinnGen GWAS
Matrix metalloproteinase-1	/	/	21,758	2020	12,705,782	European	Folkersen L et al.
Matrix metalloproteinase-2	/	/	1,000	2019	501,428	European	Suhre K et al.
Matrix metalloproteinase-3	/	/	21,758	2020	13,057,986	European	Folkersen L et al.
Matrix metalloproteinase-7	/	/	21,758	2020	12,938,892	European	Folkersen L et al.
Matrix metalloproteinase-8	/	/	1,000	2019	501,428	European	Suhre K et al.
Matrix metalloproteinase-9	/	/	3,301	2018	10,534,735	European	Sun BB et al.
Matrix metalloproteinase-10	/	/	21,758	2020	12,718,221	European	Folkersen L et al.
Matrix metalloproteinase-12	/	/	21,758	2020	13,100,092	European	Folkersen L et al.
Matrix metalloproteinase-13	/	/	1000	2019	501,428	European	Suhre K et al.
Matrix metalloproteinase-14	/	/	3,301	2018	10,534,735	European	Sun BB et al.
Matrix metalloproteinase-16	/	/	3,301	2018	10,534,735	European	Sun BB et al.
Matrix metalloproteinase-17	/	/	3,301	2018	10,534,735	European	Sun BB et al.

GWAS, genome-wide association study

ruptured intracranial aneurysms ($OR_{IVW} = 1.533$; 95% CI 1.190–1.975; $P_{IVW} = 9.41 \times 10^{-4}$). MR-IVW did not present heterogeneity. The MR-PRESSO test showed no horizontal pleiotropy. This causal association persisted after Bonferroni multiple correction. In addition, when considering intracranial aneurysm as an exposure factor and atherosclerosis as an outcome, results indicate that ruptured intracranial aneurysms are a risk factor for cerebral atherosclerosis ($OR_{MR-Egger} = 4.203$; 95% CI 1.293–13.658; $P_{MR-Egger} = 0.034$). However, the p-value did not pass multiple testing corrections, and the results exhibit pleiotropy (Pleiotropy P Value <0.05). The summary results and sensitivity analyses of the two-sample analysis are shown in eTable 6. The genetic instruments used for dual-sample MR analysis are shown in eTables 2 and 3.

Bidirectional Mendelian Randomization (Atherosclerosis/Intracranial Aneurysm and Matrix Metalloproteinases)

After establishing the causal relationship between atherosclerosis and intracranial aneurysms, we performed a two-way MR analysis to investigate the genetic predictive association between matrix metalloproteinase levels and the target diseases and to determine whether matrix metalloproteinases act as a cause or a result of the diseases. **Figure 3** displays the results with P values less than 0.05. Our analysis revealed a genetically elevated risk of peripheral atherosclerosis, resulting in elevated circulating MMP10 levels ($\beta_{IVW} = 0.087$; 95% CI 0.007 to 0.167; $P_{IVW} = 0.033$), MMP12 levels ($\beta_{IVW} = 0.083$; 95% CI 0.022 to 0.145; $P_{IVW} = 0.008$) and MMP13 levels ($\beta_{IVW} = 0.399$; 95% CI 0.063 to 0.735; $P_{IVW} = 0.020$). A genetically predicted elevated risk for coronary atherosclerosis was associated with increased circulating MMP14 levels ($\beta_{IVW} = 0.141$; 95% CI 0.050 to 0.232; $P_{IVW} = 0.002$). Unruptured brain aneurysms reduce circulating MMP2 levels ($\beta_{MR-Egger} = -0.425$; 95% CI -0.788 to -0.062; $P_{MR-Egger} = 0.041$) but increase MMP13 levels ($\beta_{IVW} = 0.116$; 95% CI 0.001 to 0.230; $P_{IVW} = 0.048$). Other atherosclerotic risks increase MMP10 levels ($\beta_{IVW} = 0.078$; 95% CI 0.011 to 0.145; $P_{IVW} = 0.022$). MR-IVW and MR Egger did not present heterogeneity. The MR-PRESSO test showed no horizontal pleiotropy.

Next, we analyzed the impact of circulating MMPs levels on genetic susceptibility to the two target diseases. After investigating 12 MMPs, we discovered that higher circulating MMP12 levels were correlated with increased risks of unruptured intracranial aneurysms ($OR_{IVW} = 1.136$; 95% CI 1.031–1.252; $P_{IVW} = 0.010$) and ruptured intracranial arteries ($OR_{simple\ mode} = 1.217$; 95% CI 1.016–1.458; $P_{simple\ mode} = 0.038$), while they were associated with a reduced risk of peripheral atherosclerosis ($OR_{IVW} = 0.883$; 95% CI 0.846–0.922; $P_{IVW} = 1.47 \times 10^{-8}$) and other atherosclerosis ($OR_{IVW} = 0.893$; 95% CI 0.855–0.933; $P_{IVW} = 4.62 \times 10^{-7}$). Additionally, elevated levels of circulating MMP10 were linked to a heightened risk of cerebral atherosclerosis ($OR_{MR-Egger} = 3.398$; 95% CI 1.183–9.757; $P_{MR-Egger} = 0.033$), while higher levels of circulating MMP8 were related to a decreased risk of coronary atherosclerosis ($OR_{MR-Egger} = 0.933$; 95% CI 0.886–0.983; $P_{MR-Egger} = 0.021$). And higher levels of circulating MMP1 were related to a decreased risk of other atherosclerosis ($OR_{IVW} = 0.948$; 95% CI 0.899–0.999; $P_{IVW} = 0.045$). MR-IVW and MR Egger did not present heterogeneity. The MR-PRESSO test showed no horizontal pleiotropy. The summary results and sensitivity analyses of the two-sample analysis are shown in eTable 7 and 8. The genetic instruments used for dual-sample MR analysis are shown in eTables 4 and 5.

Two-way MR analysis of plasma matrix proteases and intracranial aneurysm and atherosclerotic disease, as described earlier, revealed a reciprocal correlation between MMP12 levels and the risk of peripheral atherosclerosis. Although the above causal relationships disappear after Bonferroni multiple correction, they still have certain reference values.

Figure 2.

Two-way MR analysis of the associations between atherosclerosis and intracranial aneurysms ($P < 0.05$).

CERAS, cerebral atherosclerosis; PAS, peripheral atherosclerosis; IAH, ruptured intracranial aneurysm; IAN, unruptured intracranial aneurysm.

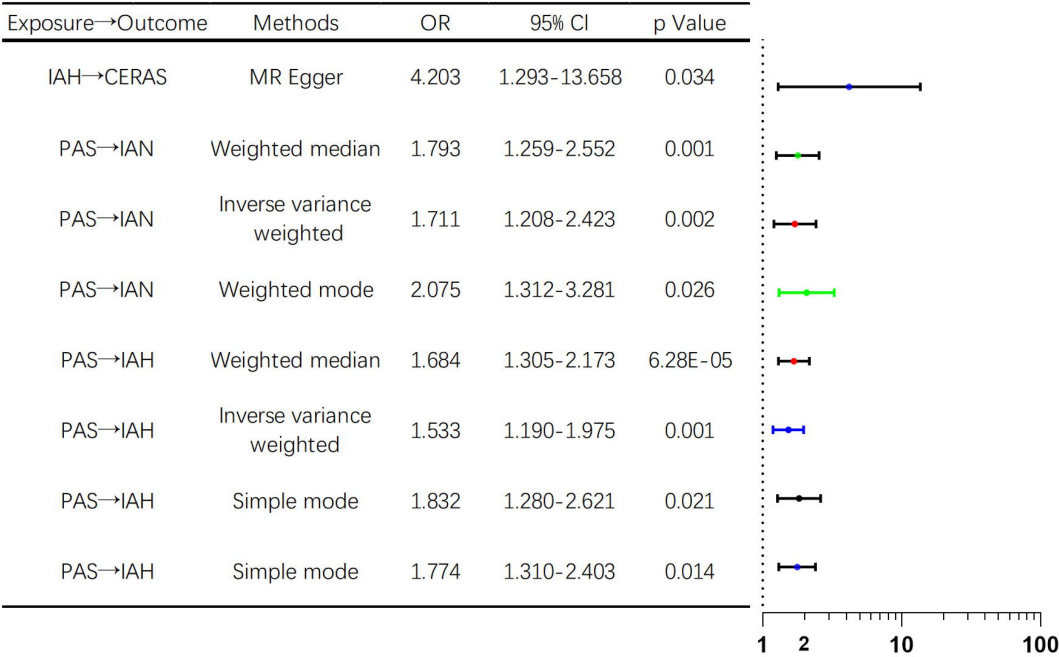
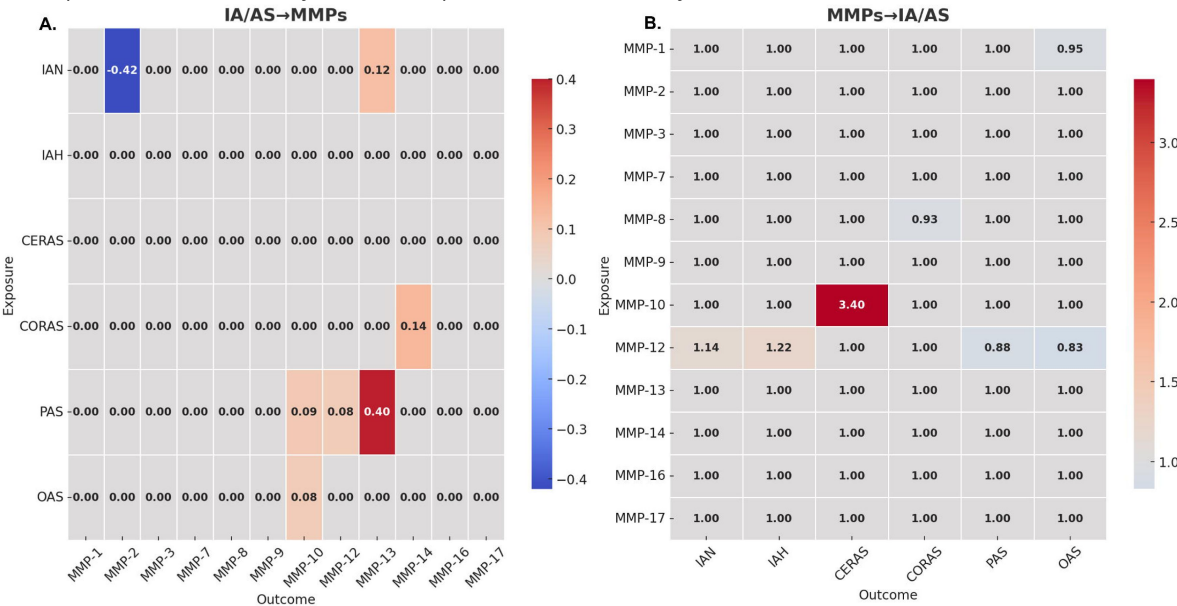


Figure 3.

Two-way MR analysis of the associations between circulating MMPs levels and atherosclerosis and intracranial aneurysms ($P < 0.05$).

CORAS, coronary atherosclerosis; CERAS, cerebral atherosclerosis; PAS, peripheral atherosclerosis; OAS, other atherosclerosis; IAH, ruptured intracranial aneurysm; IAN, unruptured intracranial aneurysm.



Plasma ELISA verification

To verify the common key pathogenic factor of atherosclerosis and intracranial aneurysms identified through MR, specifically the role of plasma MMP12 in these conditions, we collected samples from patients with carotid atherosclerosis and intracranial aneurysms, as well as from a control group of healthy participants. Plasma samples were collected and measured using ELISA. The clinical baseline characteristics and plasma MMP12 levels for carotid artery stenosis patients, intracranial aneurysm patients, and normal controls are presented in **Table 3**. Compared to healthy controls, patients with aneurysms and carotid stenosis were older and had greater systolic blood pressure. Plasma MMP12 levels were significantly greater in patients with carotid atherosclerosis than in controls. Although plasma MMP12 levels in patients with intracranial aneurysms were greater than those in the control group, the difference was not statistically significant. To further control for confounding factors, we conducted a multivariate logistic regression analysis. The results indicated that plasma MMP12 levels (CAS: OR = 2.284, 95% CI 1.357-4.638, $P = 0.007$; IA: OR = 1.557, 95% CI 1.067-2.426, $P = 0.032$), age group (CAS: OR = 21.473, 95% CI 3.622-199.404, $P = 0.002$; IA: OR = 5.933, 95% CI 1.711-23.742, $P = 0.007$), and systolic blood pressure (CAS: OR = 1.129, 95% CI 1.022-1.274, $P = 0.028$; IA: OR = 1.099, 95% CI 1.033-1.185, $P = 0.006$) are risk factors for carotid artery stenosis and intracranial aneurysms. The regression analysis results are shown in **Table 4**.

MMP12 immunohistochemical staining

To investigate the local presence of MMP12 in atherosclerotic and aneurysmal tissues, we conducted immunohistochemical staining of MMP12 in normal fibrous tissue distal to human carotid plaques, carotid plaques, aneurysms, and normal carotid plaques in model mice. Additionally, mouse cerebral blood vessels were fluorescently stained for MMP12. Our findings revealed clear MMP12 positivity in carotid plaque tissue, while no positive staining was observed in distal normal fibrous tissue. The plaque staining results are shown in **Figure 4A**. In model mice, MMP12 was found to be distributed in the elastin layer of aneurysm tissue but not in the cerebral blood vessels of normal mice. The staining results of intracranial aneurysms are shown in **Figure 4B**. These results indicate the widespread presence of MMP12 in the local tissues of arterial plaques and aneurysms, suggesting its potential involvement in the pathogenesis of atherosclerosis and aneurysms and confirming our initial hypothesis.

Discussion

The cross-sectional investigation revealed that patients with intracranial aneurysms have a greater incidence of atherosclerosis. Two-sample MR analysis indicated an association between peripheral atherosclerosis and the risk of intracranial aneurysms. In view of the role of matrix metalloproteinases in atherosclerosis and aneurysms in previous literature^{12,13}, we performed a bidirectional MR analysis on the relationships between MMPs and both atherosclerosis and intracranial aneurysms. Mendelian randomization results indicate that an increased prevalence of peripheral arterial atherosclerosis is associated with elevated circulating MMP12 levels. Additionally, elevated circulating MMP12 is associated with an increased risk of intracranial aneurysm. The results of subsequent ELISA and specimen validation were consistent with the findings from Mendelian randomization. Overall, our research confirms a causal link between atherosclerosis and the development of intracranial aneurysms, identifying and validating MMP12 as a relevant biomarker.

The relationship between atherosclerosis and aneurysm development is complex and multifaceted. Both conditions share common risk factors, including smoking, hypertension, and hypercholesterolemia¹⁴. Although atherosclerotic plaques are often found in aneurysmal

Table 3.

Baseline table of patients with atherosclerosis and intracranial aneurysm serum.

Disease	Variables	All participants (n=79)	HC (n=35)	IA (n=44)	P value	Disease	Variables	All participants (n=73)	HC (n=35)	CAS (n=38)	P value
intracranial aneurysm	age>40	53(67.089)	16(45.714)	37(84.091)	<0.001	Carotid artery stenosis	age>50	45(61.644)	10(28.571)	35(92.105)	<0.001
	sex/female	42(53.165)	17(48.571)	25(56.818)	0.466		sex/female	21(28.767)	17(48.571)	4(10.526)	<0.001
	BMI, kg/m2	24.526±3.991	23.388±3.732	25.431±3.958	0.024		BMI, kg/m2	23.875±3.538	23.388±3.732	24.324±3.287	0.265
	SBP/mmHg	130.000[124.000,138.000]	126.000[115.000,130.000]	135.000[130.000,140.000]	<0.001		SBP/mmHg	130.000[122.000,136.000]	126.000[115.000,130.000]	133.000[130.000,138.000]	0.001
	DBP/mmHg	80.709±8.748	78.886±7.678	82.159±9.261	0.101		DBP/mmHg	80.000[75.000,83.000]	79.000[75.000,81.000]	80.000[76.000,85.000]	0.516
	Heart Rate	78.101±9.703	76.657±9.905	79.250±9.381	0.244		Heart Rate	78.000[72.000,80.000]	76.000[70.000,80.000]	80.000[76.000,82.000]	0.049
	Medical history						Medical history				
	coronary heart disease	4(5.063)	0(0.000)	4(9.091)	NA		coronary heart disease	6(8.219)	0(0.000)	6(15.789)	NA
	Hyperlipidemia	3(3.797)	0(0.000)	3(6.818)	NA		Hyperlipidemia	7(9.589)	0(0.000)	7(18.421)	NA
	diabetes	5(6.329)	0(0.000)	5(11.364)	NA		diabetes	14(19.178)	0(0.000)	14(36.842)	NA
	hypertension	26(32.911)	0(0.000)	26(59.091)	NA		hypertension	27(36.986)	0(0.000)	27(71.053)	NA
	Cigarette smoking	7(8.861)	0(0.000)	7(15.909)	NA		Cigarette smoking	19(26.027)	0(0.000)	19(50.000)	NA
	Alcohol drinking	7(8.861)	0(0.000)	7(15.909)	NA		Alcohol drinking	16(21.918)	0(0.000)	16(42.105)	NA
	MMP12	5.769[4.908,6.508]	5.182[3.816,6.562]	5.969[5.285,6.508]	0.056		MMP12	6.439±2.041	5.382±1.820	7.413±1.725	<0.001

HC, healthy control; IA, intracranial aneurysm; CAS, carotid artery stenosis; BMI, body mass index; SBP, systolic blood pressure; DBP, diastolic blood pressure.

Table 4.

Logistic regression analysis of the associations between serum MMP12 levels and atherosclerosis and intracranial aneurysms.

Disease	Variables	Estimate	SE	Z	P value	Odds Ratio	Lower	Upper
intracranial aneurysm	age>40	1.781	0.661	2.695	0.007	5.933	1.711	23.742
	sex/female	0.39	0.629	0.62	0.535	1.477	0.429	5.219
	BMI, kg/m2	0.122	0.076	1.599	0.11	1.129	0.979	1.325
	SBP/mmHg	0.095	0.035	2.729	0.006	1.099	1.033	1.185
	DBP/mmHg	-0.035	0.042	-0.823	0.41	0.966	0.886	1.048
	Heart Rate	0.041	0.034	1.198	0.231	1.042	0.976	1.119
	MMP12	0.443	0.206	2.144	0.032	1.557	1.067	2.426
Carotid artery stenosis	age>50	3.067	0.998	3.074	0.002	21.473	3.622	199.404
	sex/female	-1.327	0.997	-1.33	0.183	0.265	0.032	1.778
	BMI, kg/m2	-0.028	0.146	-0.192	0.847	0.972	0.732	1.306
	SBP/mmHg	0.121	0.055	2.196	0.028	1.129	1.022	1.274
	DBP/mmHg	-0.033	0.069	-0.483	0.629	0.967	0.836	1.105
	Heart Rate	0.08	0.053	1.518	0.129	1.084	0.98	1.213
	MMP12	0.826	0.306	2.702	0.007	2.284	1.357	4.638

BMI, body mass index; SBP, systolic blood pressure; DBP, diastolic blood pressure.

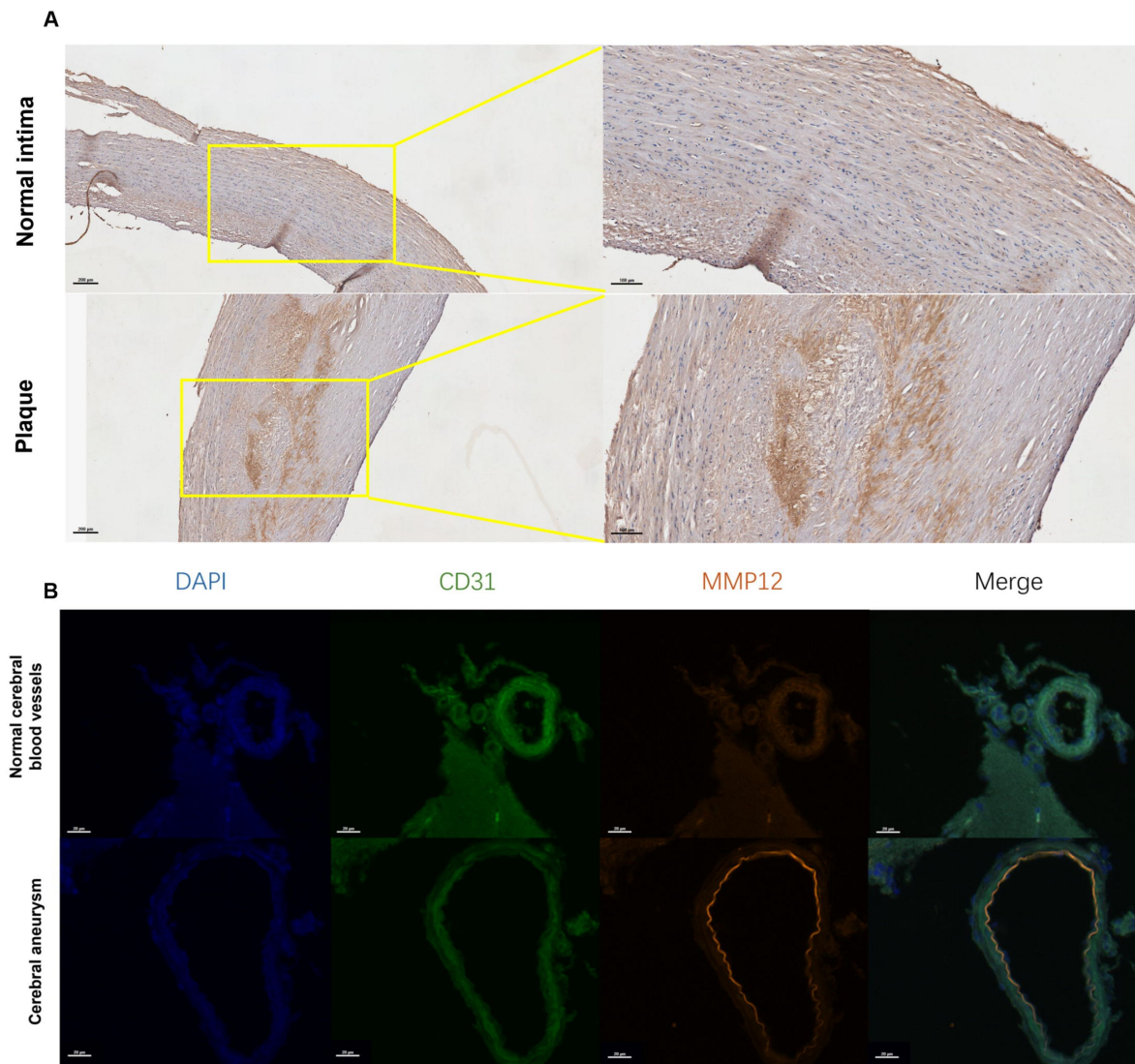


Figure 4.

A. Immunohistochemical staining of MMP12 in the necrotic area in the core of the carotid plaque and normal fibrous tissue in the distal region; B. MMP12 immunofluorescence staining of cerebral blood vessels in intracranial aneurysm model mice and normal mice.

walls⁷, it remains unclear whether atherosclerosis directly causes aneurysms. One theory suggests that aneurysms are a pathological response to atherosclerosis¹⁵. The most compelling arguments focus on arterial remodeling, with extensive *in vitro*, animal, and histological data indicating that compensatory changes occur in response to lumen narrowing and shear stress, promoting arterial expansion and potentially leading to severe medial thinning¹⁶. Proinflammatory cytokines from atherosclerotic or thrombotic inflammation combined with elastin degradation stimulated by medial proteolysis may trigger a chronic inflammatory response¹⁷. Another theory posits that atherosclerosis and aneurysm development are independent but parallel processes influenced by common environmental and genetic risk factors. A third perspective suggests that either condition can precede or stimulate the other¹⁸. Johnsen et al. analyzed data from the Tromsø Study and found no consistent dose-response relationship between the extent of atherosclerosis and abdominal aortic aneurysm size, implying simultaneous development rather than a direct causative link¹⁹. However, regional factors such as hemodynamics influence severity, suggesting that the study's cross-sectional design limits its ability to assess disease onset relationships prospectively.

Several studies have also analyzed the association between atherosclerosis and the rupture of intracranial aneurysms. The results indicate that the quantity and severity of cerebral arterial atherosclerosis are significantly correlated with the risk of aneurysm rupture²⁰. Additionally, the intima thickness of carotid plaques may serve as a potential predictor for aneurysm rupture²¹. Therefore, it can be inferred that atherosclerosis may be associated with the progression of intracranial aneurysms.

In summary, previous research has predominantly relied on retrospective analyses exploring indirect associations between these two conditions, resulting in ambiguous and uncertain conclusions. No previous studies have used MR to analyze their relationships, making our research the first to provide a clear answer on causal associations from a genetic perspective using this method. Furthermore, to explore the mechanisms underlying atherosclerosis-induced intracranial aneurysms, we identified MMP12 as a key mediator through MR and validated its role through serological and *in situ* specimen analyses. MMP12, a zinc-dependent endopeptidase involved in extracellular matrix (ECM) degradation, is secreted primarily by macrophages and is crucial for tissue remodeling, inflammation, and various pathological conditions^{22,23}. Notably, MMP12 is significantly upregulated in atherosclerotic lesions^{24,25}. During the development of atherosclerosis, damage to the vascular wall triggers an inflammatory response, leading to the aggregation of inflammatory cells such as macrophages, which secrete MMP12²⁶. Additionally, in atherosclerotic lesions, activated vascular smooth muscle cells also produce MMP12, resulting in elevated MMP12 levels²⁷. Elevated MMP12 levels have been linked to abdominal aortic aneurysms, promoting ECM degradation and arterial wall weakening²⁸. Studies using MMP12 knockout mice demonstrated that MMP12 deficiency significantly impaired aneurysm growth and reduced elastin fiber degradation, underscoring its critical role in aneurysm development and progression²⁹. Although the direct role of MMP12 in intracranial aneurysms has not been well established, similar ECM degradation and inflammatory mechanisms may be involved. The ability of MMP12 to degrade elastin and other structural proteins may weaken arterial walls, potentially facilitating intracranial aneurysm formation and rupture²². While previous research has reported the roles of MMP12 in atherosclerosis and aneurysms, no study has linked MMP12 to both conditions. Our study confirmed that atherosclerosis induces elevated plasma MMP12, which promotes the development of intracranial aneurysms.

Our study has limitations. First, the cross-sectional data were derived from a single center and relied on keyword searches within the imaging system, potentially introducing selection bias. Additionally, the baseline information included only age and sex and lacked information on other comorbidities and risk factors, such as smoking and drinking, preventing us from excluding the influence of these traditional risk factors on the prevalence of atherosclerosis and intracranial aneurysms. Second, in our analysis of the association between MMPs and the two diseases, the

causal relationship derived from our MR study lost significance after Bonferroni correction, suggesting a weaker association and the potential existence of more meaningful biomarkers. However, this result was validated through serological and in situ specimen experiments, indicating that the identified biomarker still holds reference value. Third, all MR analyses were based on individuals of European descent, limiting the applicability of the results to Asian populations. Although the serological and atherosclerotic plaque data were derived from Asian populations and aligned with the MR findings, further studies with larger sample sizes are needed to confirm generalizability. Fourth, this study did not conduct prospective analyses of the real-world association between atherosclerosis and intracranial aneurysms, necessitating further large-scale cohort studies to observe incidence rates. Finally, in our Mendelian randomization analysis, we observed a significant promoting effect of peripheral atherosclerosis on intracranial aneurysm formation. However, no significant association was found between intracranial atherosclerosis and intracranial aneurysms. It is important to note that the results of Mendelian randomization are intended to provide clues, and they are specific to the data used in this study. Negative results only suggest that no significant genetic association was found, and do not imply that they do not play a role in disease development in the real world. Therefore, further research is needed to explore the impact of atherosclerosis at different sites on intracranial aneurysms.

Conclusion

This study utilized a large-scale cross-sectional dataset to investigate the relationship between atherosclerosis and the incidence of intracranial aneurysms. Through MR analysis, we explored causal relationships and identified intermediate biomarkers, ultimately validating these biomarkers through serological and in situ specimen immunostaining experiments. Our research elucidates the causal link between atherosclerosis and intracranial aneurysms by identifying and verifying MMP12 as a critical biomarker, thereby providing new methods and evidence to explain the association between these two conditions.

Data Availability

The raw data supporting the conclusions of this article will be made available by the authors without undue reservation. All data used for Mendelian randomization analyses are publicly available from the respective GWAS.

Additional information

Ethics statement

The studies involving human participants were reviewed and approved by Beijing Tiantan Hospital, Capital Medical University (Ethics approval number: KY2022-051-02). The patients/participants provided written informed consent to participate in this study.

All the animals were acclimated under standard laboratory conditions (ventilated room, 25 ± 1 °C, 60 ± 5 % humidity, 12 h light/dark cycle) and had free access to standard water and food (SYXK-2024-0003). All procedures were conducted in accordance with the “Guiding Principles in the Care and Use of Animals” (China) and were approved by the Laboratory Animal Ethics Committee of Beijing Neurosurgical Institute (BNI202201011).

Funding

This study was supported by the National Key Research and Development Program of China (2021YFC2500502) and the National Natural Science Foundation of China (82301451).

Authors' contributions

Dong Zhang and Peicong Ge conceived the idea; Wei Liu conducted the analyses and wrote the paper; Zhaoxu Zheng developed the animal model and provided intracranial aneurysm specimens. Chenglong Liu, Yuanren Zhai, Shuang Wang and Liangran Huang provided the data; all authors contributed to the revisions.

List of abbreviations

- MMP: matrix metalloproteinase
- MR: Mendelian Randomization
- MRA: magnetic resonance angiography
- AWE: aneurysm wall enhancement
- CTA: Computed tomography angiography
- GWAS: genome-wide association study
- SNPs: single nucleotide polymorphisms
- MR-PRESSO: MR pleiotropic residual sum and outlier
- IVW: inverse-variance weighted
- ECM: extracellular matrix
- CAS: carotid atherosclerosis
- IA: intracranial aneurysm
- IAN: unruptured intracranial aneurysm
- IAH: ruptured intracranial aneurysm
- CERAS: cerebral atherosclerosis
- CORAS: coronary atherosclerosis
- PAS: peripheral atherosclerosis
- OAS: Other atherosclerosis

Clinical trial number: *NCT05485870* [🔗](#).

Additional files

supplementary figure [🔗](#)

supplementary materials [🔗](#)

eTables [🔗](#)

References

1. Keedy A (2006) **An overview of intracranial aneurysms** *McGill journal of medicine : MJM : an international forum for the advancement of medical sciences by students* **9**:141–6 [Google Scholar](#)
2. Li MH, Chen SW, Li YD, et al. (2013) **Prevalence of unruptured cerebral aneurysms in Chinese adults aged 35 to 75 years: a cross-sectional study** *Annals of internal medicine* **159**:514–21 <https://doi.org/10.7326/0003-4819-159-8-201310150-00004> | [Google Scholar](#)
3. Wang GX, Zhang D, Wang ZP, Yang LQ, Yang H, Li W (2018) **Risk factors for ruptured intracranial aneurysms** *The Indian journal of medical research* **147**:51–57 https://doi.org/10.4103/ijmr.IJMR_1665_15 | [Google Scholar](#)
4. Korja M, Lehto H, Juvela S (2014) **Lifelong rupture risk of intracranial aneurysms depends on risk factors: a prospective Finnish cohort study** *Stroke* **45**:1958–63 <https://doi.org/10.1161/STROKEAHA.114.005318> | [Google Scholar](#)
5. Brinjikji W, Zhu YQ, Lanzino G, et al. (2016) **Risk Factors for Growth of Intracranial Aneurysms: A Systematic Review and Meta-Analysis** *AJNR Am J Neuroradiol* **37**:615–20 <https://doi.org/10.3174/ajnr.A4575> | [Google Scholar](#)
6. Hokari M, Isobe M, Imai T, Chiba Y, Iwamoto N, Isu T (2014) **The impact of atherosclerotic factors on cerebral aneurysm is location dependent: aneurysms in stroke patients and healthy controls** *J Stroke Cerebrovasc Dis* **23**:2301–7 <https://doi.org/10.1016/j.jstrokecerebrovasdis.2014.04.019> | [Google Scholar](#)
7. Lu Y, Wang C, Bao Y, Huang L, Lu G, Li Y (2024) **Association between intracranial aneurysm wall enhancement and intracranial atherosclerotic plaque: a cross-sectional study using high-resolution vessel wall imaging** *Quantitative imaging in medicine and surgery* **14**:1553–1563 <https://doi.org/10.21037/qims-23-1025> | [Google Scholar](#)
8. Rostam-Alilou AA, Jarrah HR, Zolfagharian A, Bodaghi M (2022) **Fluid-structure interaction (FSI) simulation for studying the impact of atherosclerosis on hemodynamics, arterial tissue remodeling, and initiation risk of intracranial aneurysms** *Biomechanics and modeling in mechanobiology* **21**:1393–1406 <https://doi.org/10.1007/s10237-022-01597-y> | [Google Scholar](#)
9. Pierce BL, Ahsan H, Vanderweele TJ (2011) **Power and instrument strength requirements for Mendelian randomization studies using multiple genetic variants** *International journal of epidemiology* **40**:740–52 <https://doi.org/10.1093/ije/dyq151> | [Google Scholar](#)
10. Verbanck M, Chen C-Y, Neale B, Do R (2018) **Detection of widespread horizontal pleiotropy in causal relationships inferred from Mendelian randomization between complex traits and diseases** *Nature Genetics* **50**:693–698 <https://doi.org/10.1038/s41588-018-0099-7> | [Google Scholar](#)
11. Burgess S, Bowden J, Fall T, Ingelsson E, Thompson SG (2017) **Sensitivity Analyses for Robust Causal Inference from Mendelian Randomization Analyses with Multiple Genetic Variants**

Epidemiology (Cambridge, Mass) **28**:30–42 <https://doi.org/10.1097/ede.0000000000000559> | [Google Scholar](#)

12. Kim BJ, Hong EP, Youn DH, Jeon JP (2022) **Genome-Wide Association Study of the Relationship Between Matrix Metalloproteinases and Intracranial Aneurysms** *Journal of clinical neurology (Seoul, Korea)* **18**:163–170 <https://doi.org/10.3988/jcn.2022.18.2.163> | [Google Scholar](#)
13. Watanabe N, Ikeda U. (2004) **Matrix metalloproteinases and atherosclerosis** *Current atherosclerosis reports* **6**:112–20 <https://doi.org/10.1007/s11883-004-0099-1> | [Google Scholar](#)
14. Golledge J, Muller J, Daugherty A, Norman P (2006) **Abdominal aortic aneurysm: pathogenesis and implications for management** *Arterioscler Thromb Vasc Biol* **26**:2605–13 <https://doi.org/10.1161/01.ATV.0000245819.32762.cb> | [Google Scholar](#)
15. Kaschina E, Scholz H, Steckelings UM, et al. (2009) **Transition from atherosclerosis to aortic aneurysm in humans coincides with an increased expression of RAS components** *Atherosclerosis* **205**:396–403 <https://doi.org/10.1016/j.atherosclerosis.2009.01.003> | [Google Scholar](#)
16. Pasterkamp G, de Kleijn DP, Borst C (2000) **Arterial remodeling in atherosclerosis, restenosis and after alteration of blood flow: potential mechanisms and clinical implications** *Cardiovascular research* **45**:843–52 [https://doi.org/10.1016/s0008-6363\(99\)00377-6](https://doi.org/10.1016/s0008-6363(99)00377-6) | [Google Scholar](#)
17. Ward MR, Pasterkamp G, Yeung AC, Borst C. (2000) **Arterial remodeling. Mechanisms and clinical implications** *Circulation* **102**:1186–91 <https://doi.org/10.1161/01.cir.102.10.1186> | [Google Scholar](#)
18. Golledge J, Norman PE (2010) **Atherosclerosis and abdominal aortic aneurysm: cause, response, or common risk factors?** *Arterioscler Thromb Vasc Biol* **30**:1075–7 <https://doi.org/10.1161/atvbaha.110.206573> | [Google Scholar](#)
19. Johnsen SH, Forsdahl SH, Singh K, Jacobsen BK (2010) **Atherosclerosis in abdominal aortic aneurysms: a causal event or a process running in parallel? The Tromso study** *Arterioscler Thromb Vasc Biol* **30**:1263–8 <https://doi.org/10.1161/ATVBAHA.110.203588> | [Google Scholar](#)
20. Feng X, Qi P, Wang L, et al. (2019) **Relationship between cerebrovascular atherosclerotic stenosis and rupture risk of unruptured intracranial aneurysm: A single-center retrospective study** *Clin Neurol Neurosurg* **186**:105543 <https://doi.org/10.1016/j.clineuro.2019.105543> | [Google Scholar](#)
21. Lundervik M, Fromm A, Haaland Ø A, et al. (2014) **Carotid intima-media thickness--a potential predictor for rupture risk of intracranial aneurysms** *Int J Stroke* **9**:866–72 <https://doi.org/10.1111/ijvs.12159> | [Google Scholar](#)
22. Aristorena M, Gallardo-Vara E, Vican M, et al. (2019) **MMP-12, Secreted by Pro-Inflammatory Macrophages, Targets Endoglin in Human Macrophages and Endothelial Cells** *Int J Mol Sci* **20** <https://doi.org/10.3390/ijms20123107> | [Google Scholar](#)
23. Amor M, Bianco V, Buerger M, et al. (2023) **Genetic deletion of MMP12 ameliorates cardiometabolic disease by improving insulin sensitivity, systemic inflammation, and atherosclerotic features in mice** *Cardiovascular diabetology* **22**:327 <https://doi.org/10.1186/s12933-023-02064-3> | [Google Scholar](#)

24. Yamada S, Wang KY, Tanimoto A, et al. (2008) **Matrix metalloproteinase 12 accelerates the initiation of atherosclerosis and stimulates the progression of fatty streaks to fibrous plaques in transgenic rabbits** *The American journal of pathology* **172**:1419–29 <https://doi.org/10.2353/ajpath.2008.070604> | [Google Scholar](#)
25. Goncalves I, Bengtsson E, Colhoun HM, et al. (2015) **Elevated Plasma Levels of MMP-12 Are Associated With Atherosclerotic Burden and Symptomatic Cardiovascular Disease in Subjects With Type 2 Diabetes** *Arterioscler Thromb Vasc Biol* **35**:1723–31 <https://doi.org/10.1161/ATVBAHA.115.305631> | [Google Scholar](#)
26. Scholtes VP, Johnson JL, Jenkins N, et al. (2012) **Carotid atherosclerotic plaque matrix metalloproteinase-12-positive macrophage subpopulation predicts adverse outcome after endarterectomy** *J Am Heart Assoc* **1**:e001040 <https://doi.org/10.1161/jaha.112.001040> | [Google Scholar](#)
27. Hu W, Wei R, Wang L, Lu J, Liu H, Zhang W (2018) **Correlations of MMP-1, MMP-3, and MMP-12 with the degree of atherosclerosis, plaque stability and cardiovascular and cerebrovascular events** *Exp Ther Med* **15**:1994–1998 <https://doi.org/10.3892/etm.2017.5623> | [Google Scholar](#)
28. Sawada H, Daugherty A, Lu HS (2023) **Divergent Roles of Matrix Metalloproteinase 12 in Abdominal Aortic Aneurysms** *Circ Res* **132**:449–451 <https://doi.org/10.1161/CIRCRESAHA.123.322511> | [Google Scholar](#)
29. Longo GM, Buda SJ, Fiotta N, et al. (2005) **MMP-12 has a role in abdominal aortic aneurysms in mice** *Surgery* **137**:457–62 <https://doi.org/10.1016/j.surg.2004.12.004> | [Google Scholar](#)

Author information

Wei Liu

Department of Neurosurgery, Beijing Tiantan Hospital, Capital Medical University, Beijing, China, China National Clinical Research Center for Neurological Diseases, Beijing, China, Center of Stroke, Beijing Institute for Brain Disorders, Beijing, China

Zhaoxu Zheng

Department of Neurosurgery, Beijing Tiantan Hospital, Capital Medical University, Beijing, China, China National Clinical Research Center for Neurological Diseases, Beijing, China, Center of Stroke, Beijing Institute for Brain Disorders, Beijing, China

Chenglong Liu

Department of Neurosurgery, Beijing Tiantan Hospital, Capital Medical University, Beijing, China, China National Clinical Research Center for Neurological Diseases, Beijing, China, Center of Stroke, Beijing Institute for Brain Disorders, Beijing, China

Yuanren Zhai

Department of Neurosurgery, Beijing Tiantan Hospital, Capital Medical University, Beijing, China, China National Clinical Research Center for Neurological Diseases, Beijing, China, Center of Stroke, Beijing Institute for Brain Disorders, Beijing, China

Shuang Wang

Department of Neurosurgery, Beijing Tiantan Hospital, Capital Medical University, Beijing, China, China National Clinical Research Center for Neurological Diseases, Beijing, China, Center of Stroke, Beijing Institute for Brain Disorders, Beijing, China

Liangran Huang

Department of Neurosurgery, Beijing Tiantan Hospital, Capital Medical University, Beijing, China, China National Clinical Research Center for Neurological Diseases, Beijing, China, Center of Stroke, Beijing Institute for Brain Disorders, Beijing, China

Rong Wang

Department of Neurosurgery, Beijing Tiantan Hospital, Capital Medical University, Beijing, China, China National Clinical Research Center for Neurological Diseases, Beijing, China

Yan Zhang

Department of Neurosurgery, Beijing Tiantan Hospital, Capital Medical University, Beijing, China, China National Clinical Research Center for Neurological Diseases, Beijing, China, Center of Stroke, Beijing Institute for Brain Disorders, Beijing, China

Peicong Ge

Department of Neurosurgery, Beijing Tiantan Hospital, Capital Medical University, Beijing, China, China National Clinical Research Center for Neurological Diseases, Beijing, China, Center of Stroke, Beijing Institute for Brain Disorders, Beijing, China

For correspondence: gepeicong@163.com

Dong Zhang

Department of Neurosurgery, Beijing Hospital, National Center of Gerontology, Beijing, China, Institute of Geriatric Medicine, Chinese Academy of Medical Sciences, Beijing, China

For correspondence: zhangdong0660@aliyun.com

Editors

Reviewing Editor

Anurag Agrawal

Ashoka University, Sonapat, India

Senior Editor

Balram Bhargava

Indian Council of Medical Research, New Dehli, India

Reviewer #1 (Public review):

Summary:

The authors performed bidirectional two-sample Mendelian randomization using publicly available GWAS summary data to assess the directional causal association between atherosclerosis and intracranial aneurysms. They have used a similar strategy to identify the role of matrix metalloproteinases (MMP), especially MMP12, in mediating the above causal association. They finally substantiated these results by measuring and comparing the MMP12 levels in the plasma samples collected from carotid atherosclerosis and intracranial

aneurysm patients with those of healthy controls. Local tissue levels of MMP12 were also measured in experimental mouse models.

Strengths:

The authors have chosen to address an important problem that could be of interest to many researchers and clinicians in the subfield.

Weaknesses:

Mendelian Randomization (MR) is a powerful approach to explore the directional causal relationship between comorbid conditions using genetic variants as instrumental variables. The validity of causal inference derived from MR strongly depends on genetic instruments satisfying the three core assumptions- relevance, independence, and exclusion restriction. The violation of these assumptions is hard to verify in many real-world situations and may result in spurious results. Rigorous sensitivity analysis is essential to ensure the robustness of the results. The sensitivity analysis presented in the current manuscript is incomplete. The key points are as follows:

- (1) The GWAS summary datasets used by the authors for assessing the causal relationship between atherosclerosis and intracranial aneurysms were all from the FinnGen study and thus may have overlapping samples which is known to introduce bias into the causal estimates and inflate type 1 error rates.
- (2) Both atherosclerosis and aneurysms share common risk factors (mentioned by the authors as well) such as hypertension, cholesterol, diabetes, smoking, etc., which could lead to correlated pleiotropy while performing Mendelian randomization. MR-PRESSO may not effectively account for the same.
- (3) The authors explored the role of matrix metalloproteinases as intermediate biomarkers mediating the risk of atherosclerosis in the intracranial aneurysms. Separating the exposure to biomarker MR from biomarker to outcome MR limits the interpretation of the results. The effect size of the indirect effect cannot be assessed.
- (4) The scatter plots presented in Supplementary Figures 1-3 are neither cited nor discussed in the manuscript. Some of the plots show variability in the direction and magnitude of the causal estimates from MR-Egger and MR-IVW methods, indicating either masking of the causal estimates or directional pleiotropy. Discussing these results is crucial to inform the readers of the limitations of the derived causal estimates.
- (5) When there is substantial evidence available for the frequent coexistence of atherosclerosis and aneurysms, the additional value of the cross-sectional data showing the increased prevalence of atherosclerosis in patients with intracranial aneurysms without adjusting for confounding risk factors is not clear.
- (6) It is also not clear from the manuscript whether the authors are projecting the MMP12 as a shared biomarker or as a mediator between atherosclerosis and intracranial aneurysms. As also noted by the authors, assessment of plasma MMP12 levels in a cross-sectional sample is not sufficient to substantiate the role of MMP12 as an intermediate biomarker connecting atherosclerosis to the increased risk of intracranial aneurysms.

Impact:

The findings from this study can form the basis for a more systematic analysis towards identifying molecular intermediates mediating the risk of atherosclerosis in patients with intracranial aneurysms or vice versa, which in turn helps develop novel strategies to manage these comorbid conditions.

Reviewer #2 (Public review):

The manuscript by Liu and colleagues applied Mendelian Randomization (MR) techniques to study the causal relationship of atherosclerosis (categorized into four subtypes) and intracranial aneurysms (classified as unruptured or ruptured), as well as the potential mediation by 12 plasma matrix metalloproteinase (MMP) levels. The authors have followed rigorous MR analysis guidelines by using multiple analytical approaches, implementing strict selection criteria, and employing comprehensive sensitivity analyses. One of the strengths is the lack of overlapping samples in their two-sample MR analysis. This approach helps mitigate potential biases and increases the reliability of their causal inference. The analysis is fundamentally sound, but there are still several nuanced areas where the methodology could be strengthened. Given that most of the identified causal associations do not hold after correcting for multiple tests, the conclusions should be carefully reviewed to be fully supported by the results.

The recommendations below are meant to enhance the already robust approach.

- (1) The selection of 12 MMPs lacks a clear, explicit rationale in the provided excerpt. A more detailed explanation of why these specific MMPs were chosen would strengthen the methodological rigor.
- (2) Adjusting p-value for multiple testing using Bonferroni correction needs to be elucidated better.
- (3) The authors should provide a more robust explanation of why they shifted from 5×10^{-9} to 5×10^{-6} to select genomic instruments.
- (4) Egger's intercept may be a more robust approach for this study to test horizontal pleiotropy rather than MR-PRESSO.

<https://doi.org/10.7554/eLife.106521.1.sa1>

Author response:

We appreciate the constructive and thoughtful reviews provided by the reviewers and editorial team. We thank you for the opportunity to submit a provisional response and are grateful for the detailed and critical feedback that will strengthen our work. Below, we provide a summary of our planned revisions in response to the public reviews from Reviewer #1 and Reviewer #2.

Reviewer #1 – Public Review Response Plan**(1) Sample Overlap (MR Bias):**

We plan to replace several non-overlapping GWAS data sources to validate the association between aneurysms and atherosclerosis, thereby eliminating bias and Type I errors caused by sample overlap.

(2) Multivariable MR (MVMR):

We will attempt to incorporate known confounding factors (e.g., hypertension, smoking, diabetes) within the multivariable MR framework to verify the robustness of our results.

(3) Clarifications and Presentation:

- We will correct eTable citations.
- Distinguish correctly between "incidence" and "prevalence".
- Reorganize results to consistently present primary analyses first (IVW), followed by sensitivity results.
- Expand the methods section to fully reflect all analyses.

Reviewer #2 – Public Review Response Plan

(1) Justification of MMP Selection:

We will provide a detailed rationale for the inclusion of the 12 MMPs, based on prior literature and biological relevance.

(2) Multiple Testing Clarification:

We will clarify the Bonferroni correction strategy, explicitly accounting for all tests (e.g., 72 comparisons $\times$ multiple MR methods).

(3) Instrument Selection Threshold:

- We agree with the reviewer and will revise the SNP selection strategy, starting from $p < 5 \times 10^{-8}$ and only relaxing thresholds when fewer than 3 instruments are found.

- Clarify the reasons why we do not use LD proxies.

(4) Pleiotropy and Heterogeneity Tests:

- We will add Egger's intercept results alongside MR-PRESSO.

- Specify the R packages used (e.g., TwoSampleMR).

- To prevent cluttered data presentation, we have included both heterogeneity and pleiotropy p-values in the supplementary tables.

- Supplement forest plots showing outlier exclusion effects.

(5) Clarifications in Figures and Tables:

- Fix the duplicated "simple mode" entry in Figure 2.

- Correct inconsistencies in p-values between figures and text.

- Improve figure legends (e.g., color bar labels, panel identifiers).

- Revise Table 4 title for clarity.

- Remove the term "causal" where associations are nominal (e.g., $p \sim 0.05$).

<https://doi.org/10.7554/eLife.106521.1.sa0>